

Interpret inequalities



1 Write the following sentences using mathematical symbols:

- a n is greater than 17.
- b n is less than 10.
- c w is less than $\frac{2}{3}$.
- d y is greater than or equal to 12.
- e The product of 6 and x is less than or equal to -42 .
- f The sum of 5 and the product of 4 and x is greater than or equal to 20.

2 Write the following inequalities in words:

- a $p < 9$
- b $f > 14$
- c $m \leq 23$
- d $a \geq 88$

Solve inequalities

3 State the largest integer value x can have if $x \leq -2$.

4 State the smallest integer value x can have if $x > 19$.

5 State whether the following are solutions of $k \leq 15$:

- a $k = 14$
- b $k = -15$
- c $k = 15$
- d $k = 16$

6 Consider the inequality: $a + 9 < 12$.

- a Find the value of the left-hand side of the inequality when $a = 1$.
- b Is $a = 1$ a solution of $a + 9 < 12$?

7 Consider the inequality: $12 - y > 10$.

- a Find the value of the left-hand side of the inequality when $y = 0$.
- b Is $y = 0$ a solution of $12 - y > 10$?

8 Consider the inequality: $9b \geq 18$.

- a Find the value of the left-hand side of the inequality when $b = 3$.
- b Is $b = 3$ a solution of $9b \geq 18$?



9 State whether the following are solutions of $x + 2 > 6$:

a $x = 4$

b $x = 5$

c $x = 3$

d $x = -9$

10 Consider the inequality $-2x > 10$. Will the direction of the inequality symbol be reversed when solving this inequality? Explain your answer.

11 Sally attempted to solve the inequality $-18 - 6x > 30$, but has made a mistake. Her working is shown below:

Step 1 : $-18 - 6x > 30$

Step 2 : $-6x > 48$

Step 3 : $x > -8$

What was Sally's mistake?

12 Solve the following inequalities:

a $7y > 42$

b $8x \leq -48$

c $-9x < 63$

d $-4x \leq -16$

e $x - 1 < 15$

f $7 - x < 2$

g $2 \leq x - 6$

h $8x + 16 > 64$

i $7x - 49 \geq 42$

j $7x + 5 > x + 35$

k $-3(x + 9) \geq -6$

l $-9(x + 6) \geq -27$

m $23 - 3k \leq 2k + 3$

n $-4 - 4x > -7x + 2$

o $\frac{x}{3} \geq 7$

p $\frac{x}{-4} \geq 8$

q $\frac{x+9}{8} \geq 9$

r $2 - \frac{1}{2}x > 5$

s $\frac{x}{2} + \frac{x}{3} \geq 5$

t $\frac{x}{9} - \frac{x}{7} < 0$

u $\frac{a}{6} < 11$

v $\frac{a}{5} + 9 > 9$

w $91 - 4k \leq 5k + 10$

13 Write the following sentences using mathematical symbols and solve:

a The sum of 3 groups of p , and 9, is less than 24.

b The sum of 5 groups of x , and 3 is at least 23.

c Six more than the value of x is at least seven.

d Half of x is no more than five.

e Negative four groups of x is less than three.

15 Consider the following: "5 groups of p min  is no more than 10".

- a Write the sentence using mathematical symbols and solve for p .
- b Find the largest integer value of p that satisfies this condition.

16 Consider the inequality: $6(x + 6) \leq 42$.

- a Solve the inequality.
- b State whether the following are solutions of: $6(x + 6) \leq 42$.

i $x = -1$ ii $x = 2$ iii $x = 0$ iv $x = 1$

17 Consider the inequality: $30 + 3x > 30$.

- a Solve the inequality.
- b State whether the following are solutions of: $30 + 3x > 30$.

i $x = 1$ ii $x = 0$ iii $x = 3$ iv $x = 6$

18 Consider the inequality: $3(3x - 4) \geq 15$.

- a Solve the inequality.
- b State whether the following are solutions of: $3(3x - 4) \geq 15$.

i $x = 0$ ii $x = 3$ iii $x = 5$ iv $x = -3$

Applications

19 Write the following statement as an inequality in terms of T : The temperature T inside a freezer is always below -15°C .

20 Derek is saving up to buy a tablet that is selling for \$890. He has \$760 in his bank account and expects a sum of money for his birthday next month. If the amount he is to receive for his birthday is represented by x , write an inequality that models the situation where he is able to afford the tablet.

- 21** The table shows the time, t , of six swimmers in a 1500 m race. Only those who achieve a time of below 23 minutes qualify for the final. Find all the swimmers who qualify for the final.



Swimmer	Time, t (minutes)
Adam	32
Vincent	22
Yuri	28
Aaron	30
Neville	20
David	26

- 22** The table shows the distances, d , achieved by six javelin throwers. Only those who achieve a distance of more than 75 m qualify for the final. Find all the throwers who qualify for the final.

Thrower	Distance, d (metres)
Tom	63
Quentin	67
Bart	74
Bob	85
Tobias	90
Adam	71

- 23** To get a grade of C, Luke must obtain a total score of at least 300 over his four exams. So far he has taken the first three exams and achieved scores of 65, 51, and 97. If x represents what he must score on the last exam to get a C or better, write an inequality and solve for x .
- 24** Sandy has a budget for school stationery of \$46, but has already spent \$18.10 on books and folders. Let p represent the amount that Sandy can spend on other stationery. Write an inequality that shows how much she can spend on other stationery, and solve for p .
- 25** Ursula was given \$60 for a birthday present. This present, along with earnings from a Saturday job, are being set aside for a mountain bike. The job pays \$6.50 per hour, and the bike costs \$361.
- Write and solve an inequality to find the minimum number of hours that Ursula needs to work to be able to buy the mountain bike, where h represents the number of hours worked.
 - If Ursula can only work her job for a whole number of hours, find the minimum number of hours she must work to afford her bike.



- 26** Ivan wants to save up enough money so that he can buy a new sports equipment set, which costs \$60.00. Ivan has \$21.40 that he has saved from his birthday. In order to make more money, he plans to wash neighbors' windows for \$4 per window.
- a** Let w be the number of windows that Ivan washes. Write and solve an inequality to find the minimum number of windows he needs to wash in order to afford the equipment.
 - b** Find the minimum whole number of windows that Ivan must wash to be able to afford the equipment.
- 27** Charlie is saving up to buy a laptop that is selling for \$740. He has \$520 in his bank account and expects a nice sum of money for his birthday next week.
- a** If the amount he is to receive for his birthday is represented by x , write an inequality that models the situation where he is able to afford the laptop.
 - b** Would he have enough to buy the laptop if his parents were to give him \$180 for his birthday?
- 28** When breeding certain types of fish, it is recommended that the number of female fish is more than triple the number of male fish.
- a** If the number of females is represented by f and the number of males is represented by m , write an inequality that represents the recommended relationship between f and m .
 - b** Find a number of female and male fish that would satisfy the recommendation.